

Decision Problem: Dependence of Optimal Default on Channel Noise q

Abstract

We characterize how the optimal default prediction in the voluntary disclosure game depends on the channel noise parameter $q \in [0, 1]$. We show that the interval $[0, 1]$ can be partitioned into at most K sub-intervals on which a single default is optimal. A particularly clean case arises when the vanilla best constant is optimal at both endpoints: it then remains optimal for all $q \in [0, 1]$.

1 Strategic Risk Expression

For any default $i \in \{1, \dots, K\}$, the strategic risk is

$$R_S^i(q) = P(Y \neq i) + (1 - \pi_i)(1 - q)(r_X^{(-i)} - P(Y \neq i \mid U \neq i)),$$

where $\pi_i = P(U = i)$, $r_X^{(-i)}$ is the Bayes risk on the subpopulation $U \neq i$.

Each $R_S^i(q)$ is an **affine function of q** :

$$R_S^i(q) = A_i + qB_i, \quad B_i \geq 0.$$

2 Behavior at the Endpoints

Lemma 1 (Optimal Default at $q = 1$). *At $q = 1$ (complete noise), $R_S^i(1) = P(Y \neq i)$ for every i . Therefore the optimal default is always the vanilla best constant*

$$c^* = \arg \max_c P(Y = c)$$

(or any i achieving the minimal $P(Y \neq i)$).

Lemma 2 (Optimal Default at $q = 0$). *At $q = 0$ (no noise), $R_S^i(0) = P(Y \neq i) + (1 - \pi_i)\Delta_i$, where $\Delta_i \leq 0$ is the gain from features. The optimal default can be **any** of the K choices, depending on the magnitude of the gains Δ_i .*

3 Key Structural Result

Theorem 1 (Dominance When Vanilla Best Wins at Both Endpoints). *Let $i^* = c^*$ be the vanilla best constant. Suppose*

$$R_S^{i^*}(0) \leq R_S^j(0) \quad \text{and} \quad R_S^{i^*}(1) \leq R_S^j(1)$$

for all $j \neq i^*$. Then

$$R_S^{i^*}(q) \leq R_S^j(q) \quad \text{for all } q \in [0, 1] \text{ and all } j.$$

In other words, if the vanilla best default is optimal at both endpoints, it remains optimal for the entire interval $[0, 1]$.

Proof. Write $R_S^j(q) - R_S^{i^*}(q) = (A_j - A_{i^*}) + q(B_j - B_{i^*})$. By assumption the constant term $A_j - A_{i^*} \geq 0$ and the value at $q = 1$ is also non-negative. If the slope $B_j - B_{i^*} \geq 0$, the difference is non-decreasing and therefore non-negative on $[0, 1]$. If the slope is negative, the difference is non-increasing and therefore non-negative on $[0, 1]$ (its minimum on the interval is attained at the right endpoint). Hence the difference is non-negative everywhere. $\square$

This is the precise multiclass generalization of the first case in the binary Theorem 4 of the main paper.

4 General Case: Partition into At Most K Intervals

Theorem 2 (Partition of $[0, 1]$). *The interval $[0, 1]$ can be partitioned into at most K sub-intervals $[0, q_1] \cup [q_1, q_2] \cup \dots \cup [q_{m-1}, 1]$, $m \leq K$, such that on each sub-interval a single default is optimal.*

Proof. Each $R_S^i(q)$ is affine in q . The pointwise minimum of K affine functions is piecewise linear and concave, with at most $K - 1$ breaks. Hence the lower envelope consists of at most K linear pieces, each corresponding to a single optimal default. $\square$

5 Practical Algorithm to Compute the Partition

1. For every pair of defaults (i, j) with $i < j$, solve the linear equation $R_S^i(q) = R_S^j(q)$ for q .
2. Collect all solutions that lie in $(0, 1)$; these are candidate transition points.
3. Sort the distinct transition points: $0 = q_0 < q_1 < \dots < q_m < 1$.
4. For each interval $[q_{k-1}, q_k]$, evaluate R_S^i at its midpoint for all i ; the minimizing i is optimal on that interval.

Some computed crossings may lie above the lower envelope (redundancy); the midpoint evaluation automatically discards them.

6 Summary

- At $q = 1$ the optimal default is always the vanilla best constant c^* .
- At $q = 0$ any default may be optimal.
- If c^* is optimal at both endpoints, it is optimal everywhere (Theorem 3).
- In general the optimal default changes at most $K - 1$ times; the partition is obtained by the pairwise algorithm above.

This completely characterizes the q -dependence of the Stackelberg equilibrium in the decision problem.